

Homework Assignment 1 Solutions

1. IPv4 Addressing (30/100)

a) Best-fit netmask for 25 hosts and 2 routers A network with 25 hosts and 2 routers requires 27 IP addresses. The number of usable IPs in a subnet is $2^{(32 - \text{prefix})} - 2$. A /27 subnet provides $2^5 - 2 = 30$ IP addresses.

- Best-fit netmask: 255.255.255.224 or /27
- Unused IP addresses: $30 - 27 = 3$ unused addresses.

b) Division of 10.20.0.0/22 into equal-sized subnets (min 50 IPs each) Each subnet must support at least 50 usable IP addresses. The smallest subnet size that supports 50 usable IPs is a /26, which provides $2^6 - 2 = 62$ usable addresses. The original network is a /22.

- Maximum number of subnets: $2^{(26 - 22)} = 2^4 = 16$ subnets.
- Total number of usable IP addresses: $16 \times 62 = 992$ IP addresses available for hosts and routers across all subnets.

f) IANA and RIR for 194.71.64.0/22

- **RIR:** RIPE NCC
- **Organization:** Forsakringskassan
- **AS Number:** AS197942